QQ-S-351B
January 13, 1975
SUPERSEDING
Fed. Spec. QQ-S-351A
July 3, 1963

FEDERAL SPECIFICATION

SILICON, LUMP (FOUNDRY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers lump silicon for nonferrous foundry use.

1.2 Classification.

1.2.1 Grades. Lump silicon shall be supplied in the following grades, as specified:

Grade A.
Grade B.
Grade C.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

RR-S-366 - Sieves; Standard, Testing.

FSC 9630

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:05Z
Check the source to verify that this is the current version before use.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Office in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

E 31 - Methods of Chemical Analysis of Metals.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical composition. The material shall conform to the chemical requirements specified in table I.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:05Z
Check the source to verify that this is the current version before use.

Table I. Chemical composition, percent

Grade	Silicon, minimum	Aluminum, maximum	Iron, maximum	Calcium, maximum
A	97.00	1.00	1.00	0.10
B	97.00	--	1.00	.50
C	96.00	--	2.00	.50

3.2 Workmanship. The lump silicon shall be of uniform quality.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all lump silicon of the same grade submitted for inspection at one time.

4.3 Sampling. From each container of 450 pounds or less, a sample shall be made up, consisting of small pieces approximately 3/4 inch, broken-off large lumps, of halves of average-size lumps, and of small lumps and fines to a total approximate weight of 10 pounds.

4.3.1 The assorted samples taken from each container shall be combined and crushed to pass a 1/4-inch mesh sieve, after which it shall be thoroughly mixed and reduced by quartering to a composite sample of approximately 10 pounds.

4.3.2 The composite sample shall be crushed to pass a 2.380 mm. (No. 8, RR-S-366) sieve and thoroughly mixed and quartered to obtain a preliminary test sample of approximately 2 pounds. This preliminary test sample shall be thoroughly mixed and quartered again; three-quarters of this sample shall be retained and the remaining quarter shall be pulverized to pass a 100-mesh sieve and reserved as a final test sample for chemical analysis. The preliminary and final test samples of silicon shall be placed in separate rubber-stoppered bottles and labeled so as to show the name of the material, manufacturer, plant, contract or order number, lot number, and granulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.4 All acceptance tests shall be made on the final test sample representative of the lot. All collected samples of lump silicon shall be held for possible future examination should the final test sample fail to meet the requirements.

4.5 Examination.

4.5.1 Visual. Unless otherwise specified (see 6.2), samples selected in accordance with 4.3 shall be visually examined to determine compliance with the requirements for workmanship (see 3.2).

4.5.2 Preparation for shipment. Examination of the packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.6 Test procedures.

4.6.1 Chemical analysis. Except when mutually agreed upon by the supplier and purchaser, chemical analysis shall be as specified below.

4.6.1.1 Analysis for silicon content. Analysis for silicon content shall be made in accordance with ASTM E 31, sections 5 through 7.

4.6.1.2 Analysis for iron and aluminum contents. Iron and aluminum contents shall be determined as follows: Transfer a weighed sample of 2.0 to 3.0 grams to a platinum dish or large platinum crucible. Moisten with 5 to 10 ml. of water. Slowly add 25 ml. of HF and add HNO_3 dropwise until reaction ceases. Add 5 ml. of H_2SO_4 (1:1) and evaporate to dense fumes of SO_3. Cool, add sufficient water to dissolve any salts present, and transfer the solution to a pyrex beaker. Warm the solution on a hot plate and add 1 or 2 drops of HNO_3. Cool, add 5 grams of NH_4Cl, and make just ammoniacal with NH_4OH (1:1) using methyl purple indicator. Boil for 1 minute, add another drop of NH_4OH (1:1) if the indicator color changes, remove from the hot plate, and wash with hot NH_4Cl solution (20 g. NH_4Cl in 1 liter of water made slightly ammoniacal with a few drops of NH_4OH). Transfer the paper and precipitate to a weighed platinum crucible and ignite at a low temperature until the paper is charred. Finally ignite the crucible at a bright red heat. Cool and weigh the combined oxides (Fe_2O_3 and Al_2O_3). Fuse with 1 to 2 grams of $KHSO_4$, cool, leach the melt in 40 ml. of HCl (1:3), and remove the crucible. Evaporate the solution to 10 to 12 ml. and wash down the sides of the beaker with water. Heat to boiling and add $SnCl_2$ solution (5 grams $SnCl_2.H_2O$ in 10 ml. of HCl diluted to 100 ml. in water) dropwise until the yellow color is discharged and add 1 to 2 drops in excess. Dilute the reduced solution to 50 ml. and cool in a running water bath. While stirring add at one stroke 10 ml. of a saturated $HgCl_2$ solution (add 60 to 100 g. of $HgCl_2$ to 400 ml. of hot water, shake, and cool to room temperature). The resultant precipitate should be pure white in color, and in amount should be preferably not more than enough to give a pearly opalescence. If the precipitate is gray or dark, too large an excess of $SnCl_2$ has been used and it will be necessary to discard the solution and start anew. Let stand 5 minutes,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

then add 15 ml. of H_2SO_4-H_3PO_4 mixture (700 ml. of cold water, 150 ml. of H_2SO_4, and 150 ml. of H_3PO_4). Add 3 drops of sodium diphenylamine sulfonate indicator (0.1 gram in 50 ml. of water). While stirring constantly, titrate slowly with $K_2Cr_2O_7$ standard solution (0.02N or 0.05N) until the color of the solution changes from green to gray-green. Continue the titration dropwise until the first tinge of purple or violet-blue appears. Make a blank determination, following the same procedure and using the same amounts of all reagents. Calculate the percentage of iron as follows:

$$\text{Iron, percent} = \frac{(A - B)\ C \times 0.0559}{D} \times 100$$

where:

A = milliliters of $K_2Cr_2O_7$ solution required for titration of the sample,
B = milliliters of $K_2Cr_2O_7$ solution required for titration of the blank,
C = normality of the $K_2Cr_2O_7$ solution, and
D = grams of sample used.

To calculate the aluminum content, convert the percent iron to weight in grams of Fe_2O_3 (grams of Fe obtained x 1.43 = grams of Fe_2O_3). Subtract the weight of Fe_2O_3 from the total of combined weights. The difference in weight is Al_2O_3. Calculate the percentage of aluminum as follows:

$$\text{Aluminum, percent} = \frac{A \times 0.53}{B} \times 100$$

where:

A = grams of Al_2O_3, and
B = grams of sample used.

4.6.1.3 Analysis for calcium content. Calcium content shall be determined as follows: Transfer 100 grams of the sample to a 300-ml. platinum dish. Moisten with approximately 15 ml. of water. Slowly add 75 ml. of HF and add approximately 15 ml. of HNO_3 dropwise until dissolution of the sample is complete. Rinse down the walls of the platinum dish and add 25 ml. of H_2SO_4 (1:1). Evaporate to dense fumes of SO_3. Cool, and repeat the rinsing and evaporating as before. Cool, add 100 ml. of water and 10 ml. of HCl. When the salts are completely dissolved, transfer the solution to a 600-ml. beaker and dilute to approximately 400 ml. with water. Nearly neutralize the solution with NH_4OH, add 1 gram of $(NH_4)_2\ S_2O_8$, and make alkaline to litmus with NH_4OH. Boil for 5 minutes, filter, and wash with NH_4Cl solution (20 g. NH_4Cl in 1 liter of water made slightly ammoniacal with a few drops of NH_4OH). Reserve the filtrate. Dissolve the precipitate with hot HCl (1:1) into a beaker, neutralize, precipitate, and wash as above. Combine the 2 filtrates, acidify with HCl, and evaporate to approximately 200 ml. Add 0.5 grams of oxalic acid, heat to boiling, and slowly add NH_4OH (1:1) with constant stirring until the solution is slightly alkaline. Digest at 90°C. for 1 hour, cool, filter, wash with ammonium oxalate solution (1 g. per liter) and immediately afterward with water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

Transfer the paper and precipitate to a weighed crucible and ignite. Cool and weigh as CaO. Calculate the percentage of calcium as follows:

$$\text{Calcium, percent} = \frac{A \times 0.715}{B} \times 100$$

where:
A = grams of CaO, and
B = grams of sample used.

4.6.2 Reanalysis. When a lot fails to meet the chemical requirements of table I, a second portion of the preliminary test sample (see 4.3.2) shall be pulverized, and, if the second chemical analysis fails to conform to the requirements of any of the elements specified, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing (Level C). The silicon shall be packed in slack wooden barrels or steel drums. The barrels shall be provided with a bag-type barrel lining, to prevent sifting of fines and shall contain approximately 450 pounds of lump silicon.

5.2 Marking (see 6.2 and 6.3).

5.2.1 Civil agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lump silicon covered by this specification is intended for the purpose of making silicon additions to melts of nonferrous metals as follows:

6.1.1 Lump silicon, grade A. For melts of aluminum alloys where a low calcium content is a requisite.

6.1.2 Lump silicon, grades B and C. For melts of other nonferrous metals where low calcium content is not required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Grade required (see 1.2.1).
c. When workmanship requirements are not to be in accordance with 3.2 (see 4.5.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:05Z
Check the source to verify that this is the current version before use.

d. Special marking, where required (see 5.2.2).

6.3 The requirements for packing and marking for shipment (see 5.1 and 5.2), specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

MILITARY CUSTODIANS:

Army - MR
Navy - None
Air Force - 84

Review activities:

Army - None
Navy - None
Air Force - None

User activities:

Army - None
Navy - None
Air Force - None

Preparing activity:

MR

CIVILIAN AGENCIES INTEREST:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION QQ-S-351B, Silicon, Lump (Foundry)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
Army Materials and Mechanics Research Center
Watertown, Massachusetts 02172

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Director
Army Materials and Mechanics Research Center
ATTN: AMXMR-MS
Watertown, Massachusetts 02172

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.